Tribhuvan University

Faculty of Science and Technology

OFFICE OF THE DEAN

Simulation and Modeling

CSIT — 5th Semester — 2080 — Answer Sheet

Subject Code: CSC328

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Section A

1. Why model of a system is built? What is static model? Differentiate between static and dynamic mathematical models in simulation.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. What is storage in GPSS? Describe the blocks associated to storage in GPSS. A machine tool in a manufacturing shop is turning out parts at the rate of one every 5 minutes. As they are finished the parts go to an inspector who takes 10 ± 3 minutes to examine each one and rejects 10% of the parts. Represent the system in GPSS using the concept of facility and run the simulation for 500 parts.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. What are the two main properties of random numbers? Test whether the 3rd, 7th, 11th, and so on numbers in the sequence in the following random number sample are auto-correlated. (Z■ = 0.05 and Z■.■■■ = 1.96)0.12 0.01 0.23 0.28 0.89 0.31 0.64 0.28 0.83 0.93 0.99 0.15 0.33 0.35 0.91 0.41 0.60 0.27 0.75 0.88 0.68 0.49 0.05 0.43 0.95 0.58 0.19 0.36 0.69 0.87

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Section B

Attempt any Eight questions[8×5=40]

4. Explain Markov chain with suitable example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. What are steps involved in simulation study? Explain.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Generate 10 random integers using Linear congruential method where m = 1000, a = 19, c = 6, and X■ = 13.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Explain different estimation methods which are used in simulation output analysis.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Define verification and validation. Explain the process of model verification in brief.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. What is feedback system? Explain with example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. What is Calling Population? Explain Arrival and Service process in a queue.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. Explain the Monte Carlo simulation method with example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. Write short notes:a) Non-stationary Poisson Processb) Poker Test

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.